MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 05 (LINEAR EQUATIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 9(iv).
2. **Section A** comprises 3 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 1 Assertion-Reason question of 1 mark.
4. **Section C** comprises 2 Very Short Answer (VSAQ) questions of 2 marks each.
5. **Section D** comprises 1 Short Answer (3M) and 1 Long Answer / HOTS (5M) question.
6. **Section E** comprises 1 Applied Case Study with 4 sub-parts of 1 mark each.

SECTION A: Multiple Choice Questions (3 × 1 = 3 Marks)

Q1. For what value of k do the equations $3x - y + 8 = 0$ and $6x - ky = -16$ represent coincident lines? [1]

(A) 1/2 (B) $-1/2$ (C) 2 (D) -2

Q2. 8 chairs and 5 tables cost Rs. 10500, while 5 chairs and 3 tables cost Rs. 6450. The cost of each chair will be: [1]

(A) Rs. 750 (B) Rs. 600 (C) Rs. 850 (D) Rs. 900

Q3. If the sum of the ages of a father and his son is 65 years and twice the difference of their ages is 50 years, then father's age is: [1]

(A) 40 years (B) 45 years (C) 55 years (D) 65 years

SECTION B: Assertion-Reason Question (1 × 1 = 1 Mark)

Q4. **Assertion (A):** If a pair of linear equations represents coincident lines, then the system has infinitely many solutions.
Reason (R): Two coincident lines overlap completely and have every point on the line in common. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: Very Short Answer Questions (2 × 2 = 4 Marks)

Q5. Find the coordinates of the points where the line $2x - 3y = 6$ intersects the x-axis and the y-axis. [2]

Q6. Solve the following pair of linear equations for x and y: [2]

$$x + y = a + b \quad \text{and} \quad ax - by = a^2 - b^2$$

SECTION D: Short & Long Answer Questions (1 × 3 + 1 × 5 = 8 Marks)

Q7. A fraction becomes $\frac{9}{11}$, if 2 is added to both the numerator and the denominator. If 3 is added to both the numerator and the denominator, it becomes $\frac{5}{6}$. Find the fraction. [3]

Q8. Under railway reservation rules, the railway half ticket costs half the full basic fare, but the reservation charge is the same on a half ticket as on a full ticket. One reserved first class ticket from station A to station B costs Rs. 2530. Also, one reserved first class ticket and one reserved first class half ticket from A to B costs Rs. 3810. Find the full first class fare from station A to B, and also the reservation charge for a ticket. [5]

SECTION E: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q9. Case Study: Wildlife Sanctuary Census & Ecological Survey

During an educational excursion to Nandan Kannan Wildlife Sanctuary, students of Class X visited the bird sanctuary and deer park. Rohan, a keen observer, put forward a riddle to his classmates regarding the animal population. The forest ranger confirmed that the combined population of birds and deer had exactly 1000 eyes and 1400 legs. Let x denote the number of two-legged birds and y denote the number of four-legged deer.

Based on the above information, answer the following questions:

(i) Write down the linear equation representing the total count of eyes. [1]

(ii) Write down the linear equation representing the total count of legs. [1]

(iii) How many birds (x) and how many deer (y) are there in the sanctuary? [1]

(iv) Find the ratio of the number of birds to the number of deer in the sanctuary. [1]

OR

Find the total number of animals (birds + deer) in the sanctuary. [1]

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